

◆ AWS CLOUD PRACTITIONER — SESSION 9

STUDENT NOTES

CLOUD MIGRATION CONCEPTS & MODERNIZATION

◆ INTRODUCTION TO CLOUD MIGRATION

What is Cloud Migration?

Cloud migration is the process of moving:

- applications,
- databases,
- storage,
- infrastructure,
- networking,
- and workloads

from traditional on-premises environments into cloud platforms.

◆ TYPES OF CLOUD ENVIRONMENTS

Public Cloud

Cloud services shared and managed by cloud providers.

Example

AWS, Microsoft Azure, Google Cloud

Private Cloud

Cloud infrastructure dedicated to one organization.

Hybrid Cloud

Combination of on-premises infrastructure and cloud services.

Multi-Cloud

Using services from multiple cloud providers.

◆ WHY ORGANIZATIONS MIGRATE TO THE CLOUD

Major Business Drivers

Cost Optimization

Reduces hardware purchasing and maintenance costs.

Scalability

Resources can scale up or down based on demand.

Faster Innovation

Organizations deploy applications and services faster.

Reliability

Cloud providers offer highly available infrastructure.

Disaster Recovery

Cloud environments improve backup and recovery capabilities.

Global Accessibility

Applications and services can be accessed globally.

Automation

Infrastructure deployment and management can be automated.

◆ CHALLENGES OF TRADITIONAL INFRASTRUCTURE

Common Problems

Expensive Hardware

Organizations must purchase and maintain physical servers.

Limited Scalability

Scaling infrastructure can be slow and expensive.

Slow Deployment

Provisioning hardware may take days or weeks.

Downtime Risks

Hardware failures can interrupt business operations.

High Operational Overhead

Infrastructure management requires ongoing maintenance.

◆ ON-PREMISES VS CLOUD

Traditional Infrastructure	Cloud Infrastructure
Physical servers	Virtualized resources
High upfront costs	Pay-as-you-go pricing
Manual scaling	Elastic scalability
Manual operations	Automation

Traditional Infrastructure	Cloud Infrastructure
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Limited availability	Global infrastructure
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Complex disaster recovery	Built-in resilience
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◆ CLOUD MIGRATION STRATEGIES

Lift-and-Shift

Moving workloads into the cloud with minimal changes.

Advantages

- ✓ Faster migration
- ✓ Lower complexity

Limitation

- ✗ Limited cloud optimization
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Replatforming

Making small optimizations during migration.

Refactoring

Redesigning applications for cloud-native environments.

Hybrid Migration

Using both on-premises and cloud environments together.

◆ MODERNIZATION & CLOUD-NATIVE CONCEPTS

Modernization

Improving systems to better use cloud capabilities.

Cloud-Native Technologies

Containers

Portable environments for applications.

Microservices

Applications divided into smaller independent services.

Serverless Computing

Running applications without managing servers.

Managed Databases

Cloud-managed database services.

◆ BUSINESS TRANSFORMATION THROUGH CLOUD

Cloud migration helps organizations:

- improve operational efficiency,
 - increase scalability,
 - innovate faster,
 - automate services,
 - improve customer experience,
 - and operate globally.
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◆ SECURITY IN CLOUD MIGRATION

Important Security Areas

Identity & Access Management (IAM)

Controls user permissions and access.

Encryption

Protects sensitive data.

Monitoring & Logging

Tracks activity and system performance.

Backup & Recovery

Protects systems and data from failures.

Governance & Compliance

Maintains organizational standards and regulatory requirements.

◆ DISASTER RECOVERY & RESILIENCE

Cloud Advantages

- ✓ Multi-region backups
 - ✓ High availability
 - ✓ Automated failover
 - ✓ Rapid recovery
 - ✓ Reduced downtime
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◆ AWS SERVICES USED IN MIGRATION

AWS Service	Purpose
Amazon EC2	Virtual servers
Amazon S3	Object storage
Amazon RDS	Managed databases
AWS DMS	Database migration
IAM	Access management
CloudWatch	Monitoring
AWS Backup	Backup management

◆ REAL-WORLD EXAMPLE

Retail Company Migration Scenario

Problems

- slow website performance,
- downtime,
- expensive maintenance,
- limited scalability.

Cloud Solutions

- ✓ migrate applications into cloud infrastructure
 - ✓ modernize databases
 - ✓ implement auto scaling
 - ✓ improve disaster recovery
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◆ KEY TAKEAWAYS

- ✓ Cloud migration transforms businesses
- ✓ Traditional infrastructure creates operational limitations
- ✓ Cloud improves scalability and agility
- ✓ Migration strategies vary based on business needs

- ✓ Modernization improves efficiency and innovation
 - ✓ Security and governance remain critical
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◆ IMPORTANT SESSION MESSAGE

“Cloud migration is not only moving systems into the cloud.
It is transforming how organizations operate, scale, secure, and innovate.”